

Test 25: Byers-Yang theorem check

Verdict: PASS | Generated: 2026-09-02 16:39:23 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT: spectra identical for q and q+1 (?~1e-15); fractional q must DIFFER (non-neutral config).

METHOD: EXACT: spectra identical for q and q+1 (?~1e-15); fractional q must DIFFER (non-neutral config).

Test 25 HARDCORE: Byers-Yang ?<1e-10 for 3 layouts ×q?{+1,+2,?1} on 96x96 ? true; fractional q?{0.3,0.5} ?>1e-3 ? true.

Result [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 25 --no-two-pass

Independent verification guide

1. Download logs from logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[15:35:41.679] [+6456.961s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 0 vortices, model=:dirac

[15:41:02.317] [+6777.599s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

FULL COMPUTATION LOG

[15:46:12.181] [+7087.464s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

[15:51:19.675] [+7394.957s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

[15:56:20.440] [+7695.722s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 1 vortices, model=:dirac

[16:02:24.284] [+8059.566s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 1 vortices, model=:dirac

[16:07:53.533] [+8388.815s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

[16:13:03.640] [+8698.922s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

[16:18:21.308] [+9016.591s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

[16:23:36.816] [+9332.098s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

[16:29:01.269] [+9656.551s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

[16:34:17.440] [+9972.722s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=0.00, open, 4 vortices, model=:dirac

??

TEST 25 (pass 2, HARDCORE): Byers?Yang ? multi-layout / multi-q audit

3 layouts ×{+1, +2, ?1} integer rows + fractional {0.3, 0.5} rows

CONSOLE CAPTURE

??

? RUNTIME: 12 value-only diagonalizations at 96x96 (N=9216) ? expect ? 10?15 min

layout 1, q=+1: ? = 9.66e-15 < 1e-10 ?

layout 1, q=+2: ? = 1.04e-14 < 1e-10 ?

layout 1, q=-1: ? = 9.66e-15 < 1e-10 ?

layout 1, fractional q=0.3: ? = 3.35e-03 > 1e-3 ? (genuine AB effect)

layout 1, fractional q=0.5: ? = 3.74e-03 > 1e-3 ? (genuine AB effect)

layout 2, q=+1: ? = 1.24e-14 < 1e-10 ?

layout 2, q=+2: ? = 8.66e-15 < 1e-10 ?

layout 2, q=-1: ? = 1.24e-14 < 1e-10 ?

layout 3, q=+1: ? = 1.01e-14 < 1e-10 ?

layout 3, q=+2: ? = 8.89e-15 < 1e-10 ?

layout 3, q=-1: ? = 1.01e-14 < 1e-10 ?

??

integer q ?layouts PASS ? ?(q=+1,+2,?1 vs 0) < 1e-10 on 3 independent layouts

fractional ? > 1e-3 PASS ? q ? {0.3, 0.5} keep the genuine AB effect

Test 25b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS